How Far Does Translated-Query Fine-tuning Close the Arabic→English Retrieval Gap? An Empirical Study with an MTEB Arabic Benchmark and a Contamination Audit

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Abstract

Multilingual embedding models are clearly weaker when an Arabic query must find an English passage than in either monolingual direction. We measure how far fine-tuning on LLM-translated queries closes this gap and what it costs; the contributions are measurements, not a new method. (i) We assemble MTEB(ara, v1), a 19-task Arabic slice of MTEB, and audit its overlap with training data: 893 of the 1,000 MIRACL Arabic dev queries are exact copies of TyDi QA training questions. (ii) On a held-out benchmark the ar→en MRR@10 of bge-m3 rises from 63.3 to 71.2 after fine-tuning and weight interpolation, and students distilled from it gain +16.7 and +18.3 points over their off-the-shelf bases; we give the data-scaling curve and the offset between human and translated queries. (iii) A table of negative results lists what did not help. (iv) A WiSE trade-off curve shows that the gain is paid for on Arabic-only tasks: there is no knee. On MTEB(ara) the fine-tuned model is a statistical tie with bge-m3 (72.29 vs 71.98, difference +0.31, 95% task-bootstrap interval −0.34 to +0.92); two seeds trained after the recipe was frozen reproduce the test and MTEB numbers. (v) CPU deployment costs of ONNX, int8 and weight-only 8-bit variants are reported.

1 Introduction

Retrieval-augmented systems for Arabic users routinely search documents written in both Arabic and English. A multilingual bi-encoder should make this a non-issue, yet published multilingual models are markedly asymmetric: in our benchmark bge-m3 scores 87.8 for English-to-English retrieval but 63.3 when the query is Arabic and the passage English, and 67.8 in the opposite direction; multilingual-E5-base scores 49.9 and multilingual-E5-small 41.4 on ar→en. A first version of this work (v1) fine-tuned multilingual-E5-small and lifted en→ar from 61.7 to 80.4 (above bge-m3's 67.8), but left ar→en at 54.7 (mean of three seeds).

This is an empirical study. The recipe it evaluates (translated queries, mined hard negatives, a cached contrastive loss, distillation, weight interpolation) is assembled from published components, and we claim no method novelty. Our contributions are: (i) **MTEB(ara, v1) and a contamination audit**: a 19-task benchmark scored for 10 baseline models and the v2 models, and a finding that one of its retrieval tasks consists mostly of copies of a common training set; (ii) **the asymmetry and how far translated-query fine-tuning closes it**, including a data-scaling curve and the offset between machine-translated and human queries; (iii) a **negative-results table** of changes that did not help, each with its dev effect; (iv) the **WiSE trade-off curve** between cross-lingual gain and forgetting on guard tasks; (v) **deployment costs** of ONNX, int8 and weight-only 8-bit variants on CPU. We report seeds trained after the recipe was frozen as the confirmatory result, and give the pre-registration trail, with its amendments, in an appendix.

2 Related Work

Benchmarks. MTEB (Muennighoff et al., 2023) and its massively multilingual successor MMTEB (Enevoldsen et al., 2025) define the task suites in which our Arabic benchmark is expressed. Arabic retrieval is covered by MIRACL (Zhang et al., 2023), Mr.TyDi (Zhang et al., 2021), built on TyDi QA (Clark et al., 2020), and by the parallel QA sets XQuAD (Artetxe et al., 2020) and MLQA (Lewis et al., 2020). **Models.** Multilingual E5 (Wang et al., 2024, 2022) and M3-Embedding (bge-m3) (Chen et al., 2024) are the strong open multilingual encoders we start from. **Training components we reuse.** In-batch contrastive training (Henderson et al., 2017; Reimers and Gurevych, 2019) with gradient caching (Gao et al., 2021), mined hard negatives and their false-negative problem (Xiong et al., 2020; Qu et al., 2021), embedding-level distil-

lation (Hinton et al., 2015; Hofstätter et al., 2020), and interpolation of fine-tuned and original weights, WiSE-FT (Wortsman et al., 2021), which we use as a probe of the trade-off between gain and forgetting.

3 Benchmarks, Data and the Contamination Audit

Retrieval benchmark. The v1 benchmark uses only public data: TyDi QA GoldP, ARCD, SQuAD v1.1 (Rajpurkar et al., 2016; Mozannar et al., 2019) and XQuAD, split into article-disjoint train, dev and test pools. The test split has 6,823 queries over 9,817 passages; XQuAD's parallel questions give four directions (ar→ar, en→en, ar→en, en→ar) with identical information needs. Each slice is searched in the passages of its own pool in the passage language; we report MRR@10 per direction and the macro average of the four directions.

Translated queries. For cross-lingual training pairs, *questions only* are translated by a locally served LLM (Gemma-4-26B-A4B) into the other language; the passage keeps its language and stays the positive. v1 used 4,499 such pairs; v2 adds 20,999 from questions whose passages v1 never paired across languages. Test queries are never machine-translated, but the dev cross-lingual slices are, so absolute dev numbers are not comparable with test numbers. Translated XQuAD ar→en queries score 5.1–6.2 points higher than the human questions on the same 1,190 items for every model we tried, including models that never saw the translator's output: a near-constant offset, not evidence of an over-fit translation style, but a reason to read dev numbers only relatively.

MTEB(ara, v1). The benchmark has 19 tasks (retrieval, re-ranking, STS, pair classification, classification, clustering, bitext mining) with Arabic data, scored for 10 baseline models and for the v2 models by the same pipeline (Table 8); the task selection and per-task licences are documented with the benchmark.

Contamination audit. Most MIRACL Arabic dev queries are copies of TyDi QA training questions: against the full TyDi-ar training set, 893 of the 1,000 queries of MIRACL-HardNegatives are exact copies after normalisation. Counting near-duplicates of TyDi-ar and MIRACL-ar training questions as well, 895 queries are flagged and 105 are clean (for Mr.TyDi re-ranking 27 of 918). A near-duplicate detector (token Jaccard ≥ 0.8) used against v1's own training pool finds only 408, and an earlier labelling that relied on it alone flagged 814; both undercount, because short questions made of common words get no candidates, and those earlier figures are superseded. v2 drops every training question that is a near-duplicate of a query of any audited task (7 MTEB tasks) and every one whose positive passage overlaps an evaluation positive (Table 1). That filter has the same blind spot: a read-only exact-match check after training found 9 kept training rows (8 distinct questions) equal to MIRACL-HN dev queries (0.8% of the normalised queries) and 7 translated copies of them; none equal a Mr.TyDi query (0). Dropping the leaked queries changes every model's MIRACL-HN score by +0.2 to +0.3 points, including models that never trained on them, so the leak does not change any comparison; it is disclosed, and nothing was retrained.

Training queries	built	dropped	kept
Human questions (TyDi, ARCD, SQuAD)	16,419	1,173	15,246
Translated, v1	4,499	254	4,245
Translated, new in v2	20,999	1,113	19,886
Total (v2 recipe)	41,917	2,540	39,377

Table 1: Training queries of the v2 recipe before and after the contamination filter.

4 Models and Selection Protocol

We evaluate one recipe at three sizes; it is described for completeness, not as a contribution. All models minimise an InfoNCE loss over in-batch passages plus one mined hard negative per query, taken from ranks 1–30 of a baseline retriever in the same language pool, keeping only candidates at least 0.02 cosine below the positive. **v2-m3** fine-tunes bge-m3 with a cached multiple-negatives loss (Gao et al., 2021) (batch 64, mini-batch 16, scale 50, learning rate 1e-5, one epoch, maximum length 256) and blends the result with the original weights, $\theta = \alpha\,\theta_{\text{FT}} + (1-\alpha)\,\theta_{\text{bge-m3}}$; the selected α is 0.4 (60% original weights). **v2-base** and **v2-small** (e5-base, e5-small; scale 20, lr 3e-5, 2 epochs) are trained only on the KL term between the student's and the frozen v2-m3 teacher's softmax over the in-batch scores (temperature 20, weight 1, no cross-entropy term) and are the uniform soups of three seeds. Selection used DEV only; TEST was scored for final candidates. The selection rules were written down before the runs they govern and amended twice; the notebook entries are quoted with times-

tamps in Appendix A. The first rule fixed α before any guard evaluation (within 0.5 dev macro points of the raw model and 1 point of bge-m3 on a mono-Arabic mMARCO proxy (Bonifacio et al., 2021)), which selected $\alpha = 0.75$; on MTEB(ara) that candidate tied with bge-m3 (+0.06). A six-component validation suite built from non-test splits then selected the final α among 16 candidates; its best-scoring candidate (0.7612, a three-seed soup) and the pick (0.7486, the single-seed blend) differ by 0.0127, inside the registered tie band of 0.02, so the ingredients clause decided; the soup variant was never run on TEST or MTEB and is *not explored* here. The recipe was frozen at 2026-11-04 22:46:54 (UTC+3). Confidence intervals are percentile bootstraps over queries (2,000 resamples, within slices), paired when comparing systems.

5 Results

5.1 The asymmetry and how far fine-tuning closes it

Table 2 shows the test results and Table 3 the paired differences. Fine-tuning bge-m3 on the cross-lingual data and blending the weights back (v2-m3) lifts ar→en by +7.8 points (95% interval +6.7 to +9.0) and en→ar by +14.8; the macro average moves from 74.8 to 82.2. This closes the ar→en gap to English-to-English retrieval only partly: v2-m3 scores 71.2 against 89.8. The distilled students gain +16.7 (v2-base) and +18.3 (v2-small) ar→en points over the off-the-shelf models they start from, and +5.8 and +1.7 over plain fine-tuning of the same models; both remain far below the raw teacher's 73.9. v2-base is *above* bge-m3 on ar→en (+3.2, interval +1.3 to +5.2) at about half the parameters; v2-small is *not*: it is −3.6 points from bge-m3, though +4.6 above v1.

5.2 Data scaling and the translated-query offset

Table 4 shows dev results for the e5-small student as translated training queries are added. The first 2,642 new translations give +1.5 dev ar→en points, 59% of the gain obtained with the 10,568 available at that time (+2.6); the curve flattens quickly. The e5-base student gains +1.9. The last row of each block uses the contamination-filtered full set of 19,886 new translations. On the test split the same kind of extra data did not help the small students and helped bge-m3 (Table 5), so the curve should be read as saturating, not as a recipe for more translations. Dev cross-lingual queries are machine-translated and score higher than human questions (Section 3), so only the differences between rows are meaningful.

5.3 Negative results

Table 6 lists changes that did not help. Of the 13 dev comparisons none raises dev macro MRR@10 by more than +0.3 points (the largest drop is −2.4); baselines are 77.0 macro and 65.2 on ar→en for the small student, and 85.3 and 81.0 for the plain fine-tune of bge-m3 (held-out MIRACL-ar slice: 79.1). Distilling from the off-the-shelf bge-m3 does not help the small student, whereas distilling from the fine-tuned model does (Table 3): the teacher's cross-lingual skill is what is transferred. Replay of MIRACL-ar training questions, a preservation loss, freezing parts of the encoder and LoRA did not improve the held-out MIRACL-ar slice beyond the plain fine-tune; weight interpolation was the only lever that moved it. The three-seed soup at the selected α is marked †: its suite score is higher by a margin inside the tie band, and it was never run on TEST or MTEB, so we make no claim about it. Dynamic int8 export (Section 6) is included for completeness.

5.4 The WiSE trade-off curve

Interpolating the fine-tuned and original bge-m3 weights moves the model along a curve between the original model ($\alpha = 0$) and the raw fine-tune ($\alpha = 1$). Figure 1 plots test ar→en, test macro and the guard tasks against α. ar→en rises steadily from 63.3 to 73.9 while MIRACL-HN falls from 78.9 to 72.9 and Mr.TyDi re-ranking from 79.4 to 69.3; SemRel24 rises from 59.1 to 62.1. There is no knee that gives the cross-lingual gain for free. The selected α (0.4) keeps 74% of the ar→en gain and 74% of the macro gain: ar→en 71.2, MIRACL-HN 77.9 against 78.9 for bge-m3 (at the edge of the one-point tolerance, −1.0), Mr.TyDi re-ranking 76.0 against 79.4. The largest ar→en that keeps MIRACL-HN within 1 point of bge-m3 is 71.2, at $\alpha = 0.4$. **The curve is post hoc**: it was computed after α had been selected and frozen, so it describes the trade-off but did not choose the operating point.

5.5 MTEB(ara, v1) and its uncertainty

v2-m3 is numerically first on the aggregate (Table 8): 72.29 against 71.98 for bge-m3 and 71.44 for multilingual-E5-large. With 19 tasks, Table 9 asks whether that lead is real, with a bootstrap

Model	ar→ar	en→en	ar→en	en→ar	macro	95% CI (macro)
mE5-small	76.8	87.8	41.4	61.7	66.9	[66.0, 67.9]
v1 (e5-small FT, seed 13)	80.2	87.3	55.1	80.4	75.7	[74.9, 76.6]
e5-small FT, no KD	79.4	86.4	58.0	80.2	76.0	[75.2, 76.9]
v2-small	80.3	88.0	59.7	79.7	76.9	[76.1, 77.8]
mE5-base	78.0	89.8	49.9	71.9	72.4	[71.5, 73.4]
e5-base FT, no KD	78.9	86.3	60.7	81.4	76.8	[76.0, 77.7]
v2-base	82.2	89.7	66.6	82.2	80.2	[79.3, 81.0]
bge-m3	80.2	87.8	63.3	67.8	74.8	[73.9, 75.7]
bge-m3 raw fine-tune	**87.1**	**90.0**	**73.9**	**88.4**	**84.8**	[84.1, 85.6]
v2-m3	85.4	89.8	71.2	82.6	82.2	[81.4, 83.1]

Table 2: Test MRR@10 $\times 100$ per direction. Models are grouped by size (118M, 278M, 568M). Bold: best of the column. v1 is seed 13 (the three-seed mean of v1 is 54.7 on ar→en); "no KD" rows are plain fine-tunes on the same data, one epoch; v2-base and v2-small are three-seed soups; v2-m3 is the WiSE blend, the raw fine-tune its $\alpha = 1$ end.

	Δ ar→en		Δ macro	
Comparison (B vs A)	pts	95% CI	pts	95% CI
v1 vs mE5-small	+13.7	[+11.8, +15.6]	+8.8	[+8.1, +9.6]
v2-small vs mE5-small	+18.3	[+16.4, +20.2]	+10.0	[+9.3, +10.7]
v2-small vs v1	+4.6	[+3.4, +5.9]	+1.2	[+0.7, +1.7]
v2-base vs mE5-base	+16.7	[+14.7, +18.6]	+7.8	[+7.1, +8.5]
v2-base vs bge-m3	+3.2	[+1.3, +5.2]	+5.4	[+4.6, +6.1]
v2-small vs bge-m3	−3.6	[−5.7, −1.5]	+2.2	[+1.4, +2.9]
v2-m3 vs bge-m3	+7.8	[+6.7, +9.0]	+7.5	[+6.9, +8.0]
distilled vs plain FT, small	+1.7	[+0.6, +2.9]	+0.9	[+0.5, +1.4]
distilled vs plain FT, base	+5.8	[+4.4, +7.3]	+3.3	[+2.8, +3.9]
WiSE blend vs raw fine-tune	−2.7	[−3.7, −1.7]	−2.6	[−3.0, −2.2]

Table 3: Paired differences on the test split (percentage points).

DEV; training queries	new transl.	ar→en	en→ar	macro
e5-small student (no KD)				
none (v1 translations only)	0	65.2	77.9	77.0
+25% of the new translations	2,642	66.7	77.9	77.6
+first batch of new translations	3,904	66.9	79.1	77.8
+50% of the new translations	5,284	66.7	79.5	77.6
+all translations available at the time	10,568	67.8	79.7	77.8
full set, contamination-filtered	19,886	68.0	80.0	78.4
e5-base student (no KD)				
none (v1 translations only)	0	68.7	81.4	78.9
+all translations available at the time	10,568	70.6	82.1	79.3
full set, contamination-filtered	19,886	71.0	81.9	78.8

Table 4: Data-scaling curve (DEV, no distillation); "new transl." counts the new translated queries in each run's training set. Dev cross-lingual queries are machine-translated.

	ar→en		macro	
Student	partial	full	partial	full
e5-small	56.7	58.0	76.2	76.0
e5-base	59.9	60.7	77.3	76.8
bge-m3	72.6	73.9	84.2	84.8

Table 5: Test MRR@10 $\times 100$ with the translated queries available at the first runs (partial) versus the full v2 set.

Change (DEV, versus its own baseline)	Δ macro	Δ ar→en	Δ MIRACL-ar	kept
Other-language hard negative	+0.1	+0.5	–	no
Two same-language hard negatives	+0.1	+0.5	–	no
Language-homogeneous batches	−0.1	−0.1	–	no
Symmetric InfoNCE	−0.4	−0.7	–	no
KD from off-the-shelf bge-m3, w=1	0.0	+1.0	–	no
KD from off-the-shelf bge-m3, w=3	−1.0	−0.2	–	no
KD from off-the-shelf bge-m3, KD only	−2.4	−1.5	–	no
Two epochs	+0.3	+0.6	–	no
MIRACL-ar replay	−0.4	−0.2	−0.4	no
Embedding preservation to frozen bge-m3	−0.4	−0.1	−0.1	no
Frozen embeddings + bottom 12 layers	−0.3	+0.1	−0.3	no
LoRA (r=32)	+0.1	+0.5	−0.1	no
3-seed soup at the pick's α (versus the single seed)	−0.3	0.0	0.0	no†
Dynamic int8 export, v2-small / v2-base (TEST)	−2.4 / −5.4	−5.1 / −8.3	–	no

Table 6: Negative and null results, effect on dev versus the baseline of the same group (small student: no change; bge-m3: plain fine-tune; soup: the single seed); the last column says whether the change is in the final recipe. Δ MIRACL-ar: held-out slice of MIRACL-ar training questions, selection only. † within the selection rule's tie band; never run on TEST or MTEB.

over tasks and an exact sign test. **It is a statistical tie**: the difference is +0.31 (95% interval −0.34 to +0.92), with 10 wins, 8 losses and 1 tie (sign test p = 0.81); the first, α = 0.75 candidate was an even clearer tie (+0.06). The two confirmatory seeds give 72.28 and 72.27, reproducing the point estimate, not a significance. The largest gains (MLQARetrieval, +3.0; SIB200ClusteringS2S, +2.3) are offset by losses on Mr.TyDi re-ranking (−3.4) and MIRACL (−1.0). On the Arabic↔English subsets alone v2-m3 is ahead by +1.7 (+0.7 to +2.8) with 5 wins, 0 losses and 1 tie over 6 tasks (sign test p = 0.06), consistent with the held-out gains but based on very few tasks. **It is not first on those subsets**: multilingual-E5-large (560M) scores 75.3 against 75.0 for v2-m3 (bge-m3: 73.3). Among the smaller models v2-base leads: 74.2 on the cross-lingual subsets against 69.1 for mE5-base. For scale, bge-m3 against multilingual-E5-base differs by +4.1 (+2.8 to +5.4, p = 0.001), so the benchmark does resolve gaps of that size.

DEV selection set	ar→en	macro	held-out MIRACL-ar
e5-base student			
no KD, one epoch	71.0	78.8	–
KD from the raw fine-tune, w=1	74.5	81.2	–
KD w=3	75.6	81.8	73.1
KD only	76.1	82.0	73.2
KD only, two epochs	77.1	82.4	73.9
final recipe: teacher v2-m3, 3-seed soup	75.9	81.6	75.2
bge-m3 fine-tuned, WiSE blends (seed 13)			
bge-m3 (α=0)	73.5	78.2	80.5
WiSE α=0.3	78.3	82.7	80.3
WiSE α=0.4 (pick)	79.0	83.5	80.3
WiSE α=0.5	79.8	84.2	79.8
WiSE α=0.6	80.1	84.5	79.6
WiSE α=0.75	80.5	84.9	79.8
raw fine-tune (α=1)	81.0	85.3	79.1

Table 7: DEV MRR@10 ×100 for the distillation variants and the WiSE blends; the last column is the held-out MIRACL-ar slice.

The students. v2-base is ahead of mE5-base by +1.02 (−0.55 to +2.37; 15 wins, 4 losses, p = 0.02), and v2-small of mE5-small by +0.57 (−1.27 to +2.19): both intervals include zero. v2-small is +1.19 above v1 (+0.45 to +1.92), the one student comparison whose interval excludes zero. The price is on Mr.TyDi re-ranking, where the students lose −8.9 and −10.4 points against their bases, and v2-small's Mean (TaskType) (65.42) is *below* mE5-small's (65.70).

5.6 Guard tasks and the contamination analysis

Table 10 reports three guard tasks in the MTEB protocol. Relative to bge-m3, v2-m3 is +1.8 points on SemRel24, −1.0 on MIRACL-HardNegatives (77.9 vs 78.9) and −3.4 on Mr.TyDi re-ranking. Our bar was to stay within 1 point of bge-m3 on MIRACL while keeping test ar→en at 72 or above: MIRACL is at the edge of it and **Mr.TyDi re-ranking misses it**; neither replay, preservation, freezing nor LoRA closed the gap. The students lose far more: on MIRACL −2.3 (v2-base) and −4.1 (v2-small), on Mr.TyDi re-ranking −8.9 and −10.4.

Contamination does not explain the v2-m3 deficit away. Splitting the queries into flagged ones (exact or near copies of TyDi-ar or MIRACL-ar training questions) and clean ones (Table 11), v2-m3 trails bge-m3 on MIRACL by −1.4 on the 105 clean queries (interval −3.3 to +0.5) and by −1.1 on the 895 flagged ones: the same size in both, with a wide interval on the clean subset (Table 12). Every model scores lower on the clean queries than on the flagged ones, including off-the-shelf models, so the clean queries are harder rather than the gap being specific to bge-m3. On Mr.TyDi re-ranking

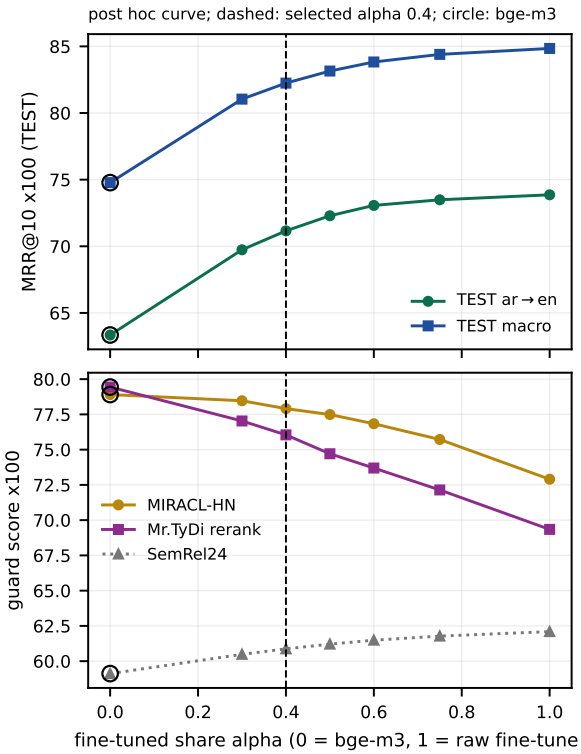


Figure 1: WiSE trade-off (post hoc): fine-tuned share α against test ar→en, test macro and the guard tasks; the circle is bge-m3, the dashed line the selected α.

the deficit is −3.2 on the clean and −7.8 on the 27 flagged queries, so the loss on that task is not a contamination effect either. The students' MIRACL deficits are larger on the clean queries (−5.7 and −7.6) than on the flagged ones (−1.8 and −3.7). The labels come from a detector with known blind spots and are best read as a lower bound on contamination; bge-m3 lists MIRACL and Mr.TyDi among its training sets, so its absolute numbers are not a clean reference either.

5.7 With a reranker on top

With the v1 fine-tuned cross-encoder over the top-50 of the first stage (Table 13), the first-stage gap of v2-m3 closes: ar→en 75.3 against 75.3 for bge-m3 (paired interval of the difference −0.7 to +0.6), macro 84.3 against 84.5. v2-m3 and bge-m3 are level once a reranker is applied; for the small models the first stage still matters (v2-base and v2-small gain +3.1 and +4.5 ar→en points over their bases after reranking). Whether a stronger first stage allows a shorter candidate list was not measured.

5.8 Confirmatory seeds

The model-selection experiments above used the first three seeds and several amended selection

Model	Task	Type	Retr.	Rerank	STS	Cls.	ar	ar-en
v2-m3	**72.3**	**70.9**	**74.4**	76.0	67.3	64.5	**70.5**	75.0
v2-m3, confirmatory seed 16	**72.3**	70.9	**74.4**	75.7	67.4	**64.5**	**70.5**	75.0
v2-m3, confirmatory seed 17	**72.3**	70.8	74.3	75.5	67.4	**64.5**	**70.4**	75.0
bge-m3	72.0	**70.9**	73.5	79.4	66.3	**64.5**	70.4	73.3
multilingual-e5-large	71.4	70.3	74.0	**79.6**	**68.1**	61.8	69.5	**75.3**
v2-base	68.9	67.6	71.1	68.2	65.5	60.4	66.3	74.2
multilingual-e5-base	67.9	67.3	68.8	77.1	64.9	59.6	66.5	69.1
v2-small	66.2	65.4	67.9	68.9	59.1	59.4	64.1	69.7
multilingual-e5-small	65.6	65.7	66.0	79.2	60.1	58.3	65.0	63.9
arabic-english-e5-small-retrieval	65.0	64.4	66.3	65.9	57.4	59.0	63.1	68.1
silma-embeddding-matryoshka-v0.1	60.3	61.9	47.1	71.4	59.2	63.2	62.2	47.9
paraphrase-multilingual-MiniLM-L12-v2	58.7	62.7	45.2	70.0	63.1	52.0	54.6	66.5
Arabic-Retrieval-v1.0	58.3	56.8	55.8	65.2	62.8	64.0	67.4	27.9
LaBSE	58.2	58.7	46.2	51.6	62.5	58.3	54.2	66.2
Arabic-Triplet-Matryoshka-V2	55.7	54.8	51.0	64.5	60.3	63.4	66.3	19.5

Table 8: MTEB(ara, v1), scores $\times 100$, sorted by Mean (Task). Task = mean over tasks, Type = mean over task types; ar / ar-en = mean over Arabic-only / Arabic–English subsets. Bold: v2 models and the best value of each column.

Mean (Task) difference, new − ref	tasks	diff	95% CI (tasks)	W/L/T	sign p
first candidate (WiSE α=0.75) vs bge-m3	19	+0.06	[−1.14, +1.13]	9/9/1	1.00
v2-m3 vs bge-m3	19	+0.31	[−0.34, +0.92]	10/8/1	0.81
ar-en cross-lingual subsets	6	+1.73	[+0.70, +2.79]	5/0/1	0.06
ar-only subsets	17	+0.09	[−0.55, +0.69]	9/8/0	1.00
v2-m3 seed 16 vs bge-m3	19	+0.30	[−0.38, +0.93]	10/8/1	0.81
v2-m3 seed 17 vs bge-m3	19	+0.29	[−0.38, +0.89]	10/7/2	0.63
v2-base vs mE5-base	19	+1.02	[−0.55, +2.37]	15/4/0	0.02
ar-en cross-lingual subsets	6	+5.14	[+2.67, +7.41]	6/0/0	0.03
ar-only subsets	17	−0.15	[−1.59, +1.10]	10/7/0	0.63
v2-small vs mE5-small	19	+0.57	[−1.27, +2.19]	11/8/0	0.65
ar-en cross-lingual subsets	6	+5.87	[+3.43, +7.94]	6/0/0	0.03
ar-only subsets	17	−0.92	[−2.56, +0.51]	6/11/0	0.33
v2-small vs v1	19	+1.19	[+0.45, +1.92]	15/4/0	0.02
ar-en cross-lingual subsets	6	+1.65	[+0.90, +2.49]	6/0/0	0.03
ar-only subsets	17	+1.04	[+0.20, +1.85]	13/4/0	0.05
reference pairs from the leaderboard of baseline models					
bge-m3 vs mE5-base	19	+4.09	[+2.83, +5.43]	17/2/0	0.001
mE5-base vs mE5-small	19	+2.26	[+0.93, +3.69]	14/5/0	0.06
v1 vs mE5-small	19	−0.62	[−2.92, +1.39]	10/9/0	1.00
bge-m3 vs mE5-large	19	+0.54	[−0.51, +1.61]	11/8/0	0.65

Table 9: Paired uncertainty of MTEB(ara) mean differences: bootstrap over tasks, wins / losses / ties (a tie is a difference below half a hundredth of a point), exact two-sided sign test. Subset rows use only the tasks that have such subsets. The lower block compares baseline models, for scale.

Model	MIRACL-HN	Mr.TyDi rerank	SemRel24
mE5-small	72.4	79.2	51.0
v1	66.3	65.9	41.2
mE5-base	73.0	77.1	59.2
bge-m3	78.9	79.4	59.1
v2-small	68.3	68.9	46.1
v2-base	70.7	68.2	59.0
v2-m3	77.9	76.0	60.9
v2-m3 − bge-m3	−1.0	−3.4	+1.8
v2-base − mE5-base	−2.3	−8.9	−0.2
v2-small − mE5-small	−4.1	−10.4	−4.8

Table 10: Guard tasks (MTEB protocol, $\times 100$): MIRACL-HardNegatives (nDCG@10), Mr.TyDi reranking (MAP), SemRel24 (Spearman).

rules, so their headline numbers are exploratory (Table 14). Two further seeds (16 and 17) were trained after the recipe was frozen (at 2026-11-04 22:46:54 (UTC+3)), with the identical arguments and the identical α; Table 15 is the confirmatory result and was reported whatever it showed. Their mean ar→en is 70.7 (macro 82.1), +7.3 points above bge-m3 (interval +6.2 to +8.4); the development seeds average 70.9 (s.d. 0.3). The MTEB(ara) means of the two seeds (72.28, 72.27) agree with the first seed's (72.29) to within a few hundredths of a point; the raw fine-tunes of the same seeds average 73.4 on ar→en.

6 Deployment Costs

We export v2-small and v2-base to ONNX (fp32, opset 17, pooling and normalisation inside the graph) and compare three ways of shrinking them for CPU serving: ONNX Runtime dynamic int8 with per-tensor or per-channel weight scales (activations quantised on the fly, cf. Zafrir et al., 2019), and weight-only 8-bit block quantisation of the MatMul weights that leaves the activations in fp32. Quality is measured as MRR@10 on the same test split against the PyTorch fp32 model, on CPU, with paired bootstrap intervals. **These measurements were made on the first distilled candidates** (single seed, teacher the raw fine-tune), not on the final soups: sizes and latencies depend only on the architecture, but the quantisation deltas are indicative for the final models and were not re-measured. Latency was measured on the 20-core ARM CPU of a shared server (not an x86 laptop or cloud VM), one process, 2 and 8 threads; numbers are therefore

Model	MIRACL-HN ar (n=105+895) all	clean	flagged	Mr.TyDi rerank (n=891+27) all	clean	flagged
bge-m3	79.0	71.9	79.9	79.4	79.1	88.5
mE5-base	73.0	67.4	73.6	77.4	77.2	82.7
mE5-small	72.4	64.7	73.3	79.3	79.2	84.1
v1	65.8	54.0	67.2	65.8	65.6	70.7
v2-m3 raw FT	72.9	63.3	74.0	69.4	69.4	69.6
v2-m3	77.9	70.5	78.8	76.0	75.9	80.7
v2-base	70.7	61.6	71.8	68.1	68.2	67.4
v2-small	68.3	57.1	69.6	68.9	68.7	74.2

Table 11: Guard tasks split by contamination: "flagged" queries are exact or near copies of a TyDi-ar or MIRACL-ar training question, "clean" queries are not. Mean per-query score in each subset, protocol of the contamination split (passage titles included).

Task	subset	n	v2-m3 − bge-m3	v2-base − mE5-base	v2-small − mE5-small
MIRACL-HN ar	clean	105	−1.4 [−3.3, +0.5]	−5.7 [−9.1, −2.2]	−7.6 [−11.3, −4.0]
MIRACL-HN ar	flagged	895	−1.1 [−1.7, −0.5]	−1.8 [−2.9, −0.7]	−3.7 [−5.0, −2.4]
Mr.TyDi rerank	clean	891	−3.2 [−4.2, −2.3]	−9.0 [−10.4, −7.6]	−10.5 [−12.0, −8.9]
Mr.TyDi rerank	flagged	27	−7.8 [−15.7, −1.5]	−15.3 [−25.0, −7.0]	−9.9 [−22.2, +2.2]

Table 12: Paired per-query differences within each subset (percentile bootstrap over queries; subsets of one hundred queries give wide intervals).

First stage (top-50)	ar→en alone	ar→en + reranker	macro alone	macro + reranker
bge-m3	63.3	75.3	74.8	84.5
mE5-base	49.9	71.3	72.4	83.5
mE5-small	41.4	68.9	66.9	82.7
v2-m3	71.2	75.3	82.2	84.3
v2-base	66.6	74.4	80.2	84.0
v2-small	59.7	73.4	76.9	83.6

Table 13: Test MRR@10 $\times 100$ with and without the unchanged v1 cross-encoder reranker over the top candidates of each first stage.

TEST, mean ± s.d. over three development seeds	ar→en	macro
raw fine-tune (α=1)	73.3 ± 0.6	84.5 ± 0.3
v2-m3 recipe (WiSE)	70.9 ± 0.3	82.2 ± 0.1
v2-base students	65.9 ± 0.2	79.9 ± 0.1
v2-small students	59.3 ± 0.3	76.8 ± 0.0

Table 14: Development seeds (test): the raw fine-tune, the recipe of the pick and the two distilled students.

Model, seed	ar→en	macro	Δ ar→en vs bge-m3	MTEB Mean (Task)
v2-m3 recipe (WiSE), seed 16	71.1	82.2	+7.7 [+6.6, +8.9]	72.28
v2-m3 recipe (WiSE), seed 17	70.3	82.0	+7.0 [+5.8, +8.1]	72.27
v2-m3 recipe (WiSE), mean of 2 seeds	70.7	82.1	+7.3 [+6.2, +8.4]	72.27
raw fine-tune (α=1), seed 16	73.6	84.7	+10.2 [+8.7, +11.8]	–
raw fine-tune (α=1), seed 17	73.1	84.3	+9.8 [+8.2, +11.3]	–
raw fine-tune (α=1), mean of 2 seeds	73.4	84.5	+10.0 [+8.5, +11.4]	–

Table 15: Confirmatory seeds, trained after the recipe was frozen (test). Intervals: paired bootstrap over queries against bge-m3 (for the mean row, of the seed-averaged per-query scores).

indicative.

Table 16 shows that the ONNX fp32 graphs reproduce the PyTorch results (0.00 and 0.00 macro points) and cut single-query latency (2 threads) from 18 to 6 ms for v2-small and from 48 to 32 ms for v2-base. Dynamic int8 with per-tensor scales is faster still (4 and 7 ms) but *not acceptable* here: −2.44 macro points for v2-small and −5.42 for v2-base (−8.27 on ar→en for the base model). Per-channel scales reduce the loss to −0.72 and −4.56, still more than we would ship for the base model. Weight-only 8-bit quantisation is lossless within noise (+0.04 and −0.11) and shrinks the models from 448 to 114 MiB and from 1059 to 270 MiB, but its CPU kernels are slower per query than fp32 (18 and 97 ms). We therefore recommend ONNX fp32 when latency matters, weight-only 8-bit when download size matters, and no dynamic int8 for these models without calibration or quantisation-aware training.

7 Data Card and Licences

Table 17 lists every dataset and model used, how it entered the study, and the licence on its Hugging Face card (checked 2026-11-04). The final recipe trains on TyDi QA (Apache-2.0), SQuAD v1.1 (CC BY-SA 4.0) and ARCD (MIT) and on machine translations of their questions produced by Gemma-4-26B-A4B (Apache-2.0); the starting weights are bge-m3 (MIT) and multilingual-E5 (MIT). MIRACL Arabic (Apache-2.0) was used only in an ablation, XQuAD (CC BY-SA 4.0) only for testing, and the remaining sources only for validation or selection. Passage text is Wikipedia content, so share-alike terms of SQuAD and Wikipedia apply to derived corpora. [TODO: licence of the released weights: MIT or CC BY-SA 4.0, the author decides; the training data include CC BY-SA sources]

Variant (CPU)	MiB	RSS	q@1 ms	p/s	MRR	Δ	Δ ar→en
v2-small PyTorch fp32	449	1818	18	14	77.4	ref.	ref.
v2-small ONNX fp32	448	2613	6	20	77.4	0.00	0.00
v2-small ONNX int8	113	2249	4	34	75.0	−2.44	−5.09
v2-small ONNX int8, per-channel	113	2258	4	33	76.7	−0.72	−1.68
v2-small ONNX weight-only 8-bit (MatMulNBits)	114	2266	18	20	77.5	+0.04	+0.05
v2-base PyTorch fp32	1061	2447	48	3	79.7	ref.	ref.
v2-base ONNX fp32	1059	3089	32	4	79.7	0.00	0.00
v2-base ONNX int8	265	2708	7	11	74.3	−5.42	−8.27
v2-base ONNX int8, per-channel	266	2736	9	11	75.1	−4.56	−6.50
v2-base ONNX weight-only 8-bit (MatMulNBits)	270	2226	97	4	79.6	−0.11	−0.18

Table 16: CPU deployment variants, measured on the first distilled candidates (see text). MiB: size on disk; RSS: peak resident memory of the benchmark process (8 threads; includes the python and data-loading baseline of that process); q@1: p50 latency of a single query (2 threads, ms); p/s: passages per second (batch 32, 8 threads); MRR: test macro MRR@10 $\times 100$; Δ: difference to PyTorch fp32 in points (macro, and ar→en).

Source	Role	Licence (Hub card)
TyDi QA GoldP (Arabic, English)	training data (final recipe)	apache-2.0
SQuAD v1.1 train	training data (final recipe)	cc-by-sa-4.0
ARCD	training data (final recipe)	mit
XQuAD	evaluation / selection only	cc-by-sa-4.0
MIRACL Arabic (queries, qrels, corpus)	training data (ablation only)	apache-2.0
Gemma-4-26B-A4B (query translator)	translated the training queries	apache-2.0
bge-m3 (base weights of v2-m3, teacher baseline)	initial weights	mit
multilingual-E5-base (base of v2-base)	initial weights	mit
multilingual-E5-small (base of v2-small)	initial weights	mit
ArSarcasm (selection suite)	evaluation / selection only	mit
XNLI (selection suite)	evaluation / selection only	no licence tag on the Hub card
Arabic STS-B (selection suite)	evaluation / selection only	apache-2.0
mMARCO Arabic (selection proxy)	evaluation / selection only	no licence tag on the Hub card
Weights of v2-m3, v2-base, v2-small	released models	[TODO: licence of the weights: author decision (training data include share-alike sources)]

Table 17: Data card: sources, roles and licences. “No licence tag” means the Hub card does not state one; nothing is assumed.

8 Limitations

Most fine-tuned numbers are single seeds; the confirmatory seeds cover the pick’s recipe only (not the students), and the soup variant of the pick was never run on TEST or MTEB. The test set is Wikipedia-style extractive QA with 1,190 parallel questions per XQuAD direction, so the intervals are wide for the cross-lingual slices; dialects, informal text and long documents are untested. Training queries are machine-translated and not human-verified; the offset check cannot rule out subtler translationese, and dev cross-lingual numbers are inflated by it. The guard analysis shows a real regression on MIRACL and Mr.TyDi, large for the students, and the clean MIRACL subset is small; the contamination labels have known blind spots and 9 kept training rows equal evaluation queries. MTEB(ara, v1) has 19 tasks; its aggregate cannot resolve differences of about a point (Table 9), and resampling tasks ignores query-level noise. The selection rules were amended during the work (Appendix A), the first candidate had been scored on MTEB before the suite existed, and the suite shares a source with one MTEB task; the post-hoc WiSE curve and the exploratory seeds should not be read as confirmatory. Deployment quality deltas were measured on earlier student candidates; latency comes from one machine type, with other jobs running, and does not include serving overhead. No RAG-level evaluation is done here, and the reranker pairing uses a reranker that was not retrained.

9 Reproducibility and Cost

Every number in this report is generated by `report/make_tables.py` from the committed result files in `results/retrieval_v2`, `results/mteb_ara` and `deploy/results`; the figure by `report/make_figures.py`; run `make -C report pdf`. The 46 training runs took 12.2 hours of training in total, the MTEB(ara) scorings of the v2 models 1.9 hours of summed task evaluation time, and the whole study about 22 hours of GPU wall-clock on a shared single-GPU machine (an upper bound on exclusive time; the translation server's time is not included). The evaluation harness is in the accompanying repository and the deployment scripts are in `deploy/`. The data are public (Table 17); the v2 model weights are [TODO: published after review: Hugging Face repository names and weights licence].

References

Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. 2020. On the cross-lingual transferability of monolingual representations. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4623–4637, Online. Association for Computational Linguistics.

Luiz Bonifacio, Vitor Jeronymo, Hugo Queiroz Abonizio, Israel Campiotti, Marzieh Fadaee, Roberto Lotufo, and Rodrigo Nogueira. 2021. mMARCO: A Multilingual Version of the MS MARCO Passage Ranking Dataset. *Preprint*, arXiv:2108.13897.

Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-embedding: Multilinguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2318–2335, Bangkok, Thailand. Association for Computational Linguistics.

Jonathan H. Clark, Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. 2020. TyDi QA: A benchmark for information-seeking question answering in typologically diverse languages. *Transactions of the Association for Computational Linguistics*, 8:454–470.

Kenneth Enevoldsen, Isaac Chung, Imene Kerboua, Márton Kardos, Ashwin Mathur, David Stap, Jay Gala, Wissam Siblini, Dominik Krzemiński, Genta Indra Winata, et al. 2025. MMTEB: Massive Multilingual Text Embedding Benchmark. *Preprint*, arXiv:2502.13595.

Luyu Gao, Yunyi Zhang, Jiawei Han, and Jamie Callan. 2021. Scaling deep contrastive learning batch size under memory limited setup. In *Proceedings of the 6th Workshop on Representation Learning for NLP (RepL4NLP-2021)*, pages 316–321, Online. Association for Computational Linguistics.

Matthew Henderson, Rami Al-Rfou, Brian Strope, Yun hsuan Sung, Laszlo Lukacs, Ruiqi Guo, Sanjiv Kumar, Balint Miklos, and Ray Kurzweil. 2017. Efficient Natural Language Response Suggestion for Smart Reply. *Preprint*, arXiv:1705.00652.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. *Preprint*, arXiv:1503.02531.

Sebastian Hofstätter, Sophia Althammer, Michael Schröder, Mete Sertkan, and Allan Hanbury. 2020. Improving Efficient Neural Ranking Models with Cross-Architecture Knowledge Distillation. *Preprint*, arXiv:2010.02666.

Patrick Lewis, Barlas Oguz, Ruty Rinott, Sebastian Riedel, and Holger Schwenk. 2020. MLQA: Evaluating cross-lingual extractive question answering. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7315–7330, Online. Association for Computational Linguistics.

Hussein Mozannar, Elie Maamary, Karl El Hajal, and Hazem Hajj. 2019. Neural Arabic question answering. In *Proceedings of the Fourth Arabic Natural Language Processing Workshop*, pages 108–118, Florence, Italy. Association for Computational Linguistics.

Niklas Muennighoff, Nouamane Tazi, Loic Magne, and Nils Reimers. 2023. MTEB: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2014–2037, Dubrovnik, Croatia. Association for Computational Linguistics.

Yingqi Qu, Yuchen Ding, Jing Liu, Kai Liu, Ruiyang Ren, Wayne Xin Zhao, Daxiang Dong, Hua Wu, and Haifeng Wang. 2021. RocketQA: An optimized training approach to dense passage retrieval for open-domain question answering. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5835–5847, Online. Association for Computational Linguistics.

Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 2383–2392, Austin, Texas. Association for Computational Linguistics.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China. Association for Computational Linguistics.

Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2022. Text Embeddings by Weakly-Supervised Contrastive Pre-training. *Preprint*, arXiv:2212.03533.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual E5 Text Embeddings: A Technical Report. *Preprint*, arXiv:2402.05672.

Mitchell Wortsman, Gabriel Ilharco, Jong Wook Kim, Mike Li, Simon Kornblith, Rebecca Roelofs, Raphael Gontijo-Lopes, Hannaneh Hajishirzi, Ali Farhadi, Hongseok Namkoong, et al. 2021. Robust fine-tuning of zero-shot models. *Preprint*, arXiv:2109.01903.

Lee Xiong, Chenyan Xiong, Ye Li, Kwok-Fung Tang, Jialin Liu, Paul Bennett, Junaid Ahmed, and Arnold Overwijk. 2020. Approximate Nearest Neighbor Negative Contrastive Learning for Dense Text Retrieval. *Preprint*, arXiv:2007.00808.

Ofir Zafrir, Guy Boudoukh, Peter Izsak, and Moshe Wasserblat. 2019. Q8BERT: Quantized 8Bit BERT. *Preprint*, arXiv:1910.06188.

Xinyu Zhang, Xueguang Ma, Peng Shi, and Jimmy Lin. 2021. Mr. TyDi: A multi-lingual benchmark for dense retrieval. In *Proceedings of the 1st Workshop on Multilingual Representation Learning*, pages 127–137, Punta Cana, Dominican Republic. Association for Computational Linguistics.

Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamalloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin. 2023. MIRACL: A multilingual retrieval dataset covering 18 diverse languages. *Transactions of the Association for Computational Linguistics*, 11:1114–1131.

A Pre-registration Trail

The selection of the v2-m3 recipe followed rules that were written down in the lab notebook of the training machine before the runs they govern. Below, the notebook entries are quoted verbatim apart from LaTeX escaping, with their timestamps: the rule that fixed α, the original replay/preservation rule, amendment 1 (a validation suite, written before any candidate was scored on it, with a same-day correction), amendment 2 (an extended range of α and a rerun of the LoRA candidate after a bug in the first run), the third pre-registration (confirmatory seeds, student rule, post-hoc curve, paired MTEB statistics), the freeze of the recipe and the start of the confirmatory seeds, and the correction of the contamination labels. The amendments changed the selection procedure after earlier selections had been made, which is why the confirmatory seeds and the post-hoc curve are reported separately.

WiSE alpha selection rule, as recorded in the results entry (2026-11-04 11:05) Selection rule fixed BEFORE any guard eval: dev macro within 0.5 of raw AND proxy within 1.0 of bge-m3 → alpha 0.75.

Original pre-registered rule (2026-11-04 14:00) 2026-11-04 ~14:00 coordinator: fix the MIRACL forgetting properly (replay + preservation), PRE-REGISTERED selection rule
- Why: v2-m3 + WiSE a.75 scores MIRACL-HN.v2 75.7 vs bge-m3 79.0 (-3.3); bar = within 1 pt while TEST ar→en >= 72.
- Replay data: MIRACL-ar TRAIN (miracl/miracl topics+qrels, miracl/miracl-corpus; Apache-2.0; judged-relevant = positives, judged-NOT-relevant = hard negatives), 500 queries HELD OUT as the selection-only slice devM (own mini-corpus of judged passages + 10k random distractors, evaluated separately, never in the dev macro). Filters: near-duplicate of any MIRACL-ar dev query / v1 test+dev query, positive overlapping a MIRACL dev positive or a v1 test passage; contam_v2 then applies the whole MTEB(ara) list. Mr.TyDi-ar train: not added (its questions ARE the TyDi-ar train questions v1/v2 already use; script-based loader).
- Preservation loss: 1 - cos(student, frozen bge-m3) on Arabic positives, weight 0.5 (sweep not affordable: one m3 run = 55 min).
- Runs (seed 13, m3 recipe of m3_b_s13): m3_r (replay), m3_p (preservation), m3_rp (both); each also WiSE a.75 / a.50 against bge-m3; evaluation = dev (4 directions), devM, mMARCO proxy. NO guard is used for selection.
- RULE (fixed now, before the runs): a candidate qualifies if (i) devM MRR@10 >= bge-m3's devM - 1.0, (ii) dev ar→en >= 79.0 (~ TEST 72), (iii) dev macro >= m3_b dev macro (85.3) - 1.5; among qualifying take the highest dev macro (ties: fewer moving parts). If none qualifies: highest devM among those with dev ar→en >= 79.0, reported as a miss.
- Seed 14 of the OLD recipe is allowed to finish (running); seed 15 is NOT run with the old recipe. KD ablations/baseline guards are queued after these runs.

Amendment 1: validation-suite selection (2026-11-04 19:05) 2026-11-04 ~19:05 - PRE-REGISTRATION: validation suite for the final v2-m3 pick (written BEFORE any candidate is scored on the new components)
Trigger: coordinator, "0.06 is a tie; the losses (Namaa, MIRACL, XNLI, AJGT) are drift away from bge-m3's general skills; select on a validation suite that mirrors the MTEB(ara)
task types, never on MTEB". Supersedes select_m3.py (kept as the record of the earlier rule). No MTEB(ara) number and no TEST number is read by this procedure.
Order of work: s15 → soup/blends → suite (interim table) → LoRA / freeze → suite (all) → selection → ONE MTEB(ara) scoring of the pick → teacher = pick → students.
DEVIATION FROM THE BRIEF (flagged to the coordinator): the brief asks for the XNLI-ar VALIDATION split "since MTEB uses test". mteb's XNLI task sets eval_splits=["test","validation"] (pylibs/mteb/tasks/pair_classification/multilingual/xnli.py) and the leaderboard number is the mean over both, so the validation split is
part of the MTEB evaluation. Instead: a seeded 3,000-pair subsample of the XNLI-ar TRAIN split (facebook/xnli, ar/train; machine-translated MNLI; entailment=1, contradiction=0, neutral dropped, as mteb does). Never trained on, never part of any MTEB split.
COMPONENTS (6, all x100, higher = better):
1 dev : v2 extended DEV macro MRR@10 over the 4 directions (ar→en, en→ar, ar→ar, en→en) [results/NAME.dev.json]
2 devM : MIRACL-ar TRAIN hold-out (500 queries, selection-only slice, excluded from training), MRR@10 [NAME.devM.json]
3 proxy : mMARCO-ar dev proxy (1,000 q / 25k docs, seed 13), MRR@10 [results/ood_dev_ar.json]
4 xnli : average precision of the cosine similarity on 3,000 XNLI-ar TRAIN pairs (seed 13) [results/suite/NAME.json]
5 cls : mean of two 8-shot-per-label logistic-regression probes (mteb protocol: 10 experiments, samples_per_label=8, max_iter=100) on ArSarcasm
(iabufarha/ar_sarcasm, licence MIT): (a) sarcasm 2-class, (b) sentiment 3-class. The probe is fitted on rows of the TRAIN split and scored on
2,000 other TRAIN rows (seeded hold-out). MTEB scores TweetSarcasmClassification on the ar_sarcasm TEST split; the TRAIN split is disjoint

from it (same source, so it is a close proxy by design). AJGT / TweetEmotion / HotelReview are not touched.
6 sts : Spearman of cosine similarity on the Arabic DEV split of stsb_multi_mt (PhilipMay/stsb_multi_mt, ar/dev, 1,500 pairs; machine-translated STSb; licence "other" - used for validation only, not redistributed). Disjoint from STS17 / STS22 / SemRel24 (the MTEB STS tasks).
NORMALISATION: per component c, x_c is min-max scaled over the POOL = {all candidates + the reference rows (bge-m3, m3_b_s13 raw, soup raw)}:
n_c = (x_c - min_pool) / max(max_pool - min_pool, 1.0 point) (the 1-point floor keeps a component whose pool range is below the noise level from being inflated)
SCORE = unweighted mean of the 6 n_c. Reported with the 6 raw components.
POOL / CANDIDATES (selectable): WiSE alpha in {0.4,0.5,0.6,0.75} of m3_b_s13 (wm_a50, wm_a75 exist; w_m3b_a40, w_m3b_a60 new);
SOUP = uniform average of the seed 13/14/15 fine-tunes (m3_b_s13/s14/s15; seed 15 is trained first), then WiSE alpha in {0.4,0.5,0.6,0.75} (soup_a40..a75);
LoRA r=32 attention+MLP lr 1e-4 (m3_lo_s13) and freeze-bottom-12 (m3_fr_s13) raw, plus their WiSE blends alpha {0.75,0.5}. 14 selectable models.
Reference rows (in the pool for normalisation, shown, NOT selectable): bge-m3, m3_b_s13 raw (alpha=1), soup raw.
ELIGIBILITY (protects the S_TIER ar→en bar): dev ar→en >= bge-m3 dev ar→en + 5.0 points.
PICK = the eligible candidate with the highest SCORE. Ties (|diff| < 0.02) → fewer ingredients: single-seed WiSE < soup WiSE < LoRA/freeze (then larger alpha).
The pick is announced here before its MTEB(ara) scoring; that scoring is run once and reported even if it is lower than final_m3_wa75_s13 (72.044). Earlier MTEB(ara) scoring of final_m3_wa75_s13 happened BEFORE this suite existed and is not used in the selection.
3-SEED REPORTING: pick is a soup → the soup is the reported model (one model; paired bootstrap over queries); otherwise the per-seed blends (13/14/15) of the picked recipe.
Students: distilled from the FINAL teacher (the pick), KD-only; teacher embeddings are recomputed from the pick (tag m3f).
Code: lab/retrieval/v2/suite_v2.py (components 4-6), suite_select.py (assembly + rule), runs/retrieval/v2/scripts/drive14.sh (queue).
- 2026-11-04 ~19:20 CORRECTION to the pre-registration above (still before any candidate has a suite number):
PhilipMay/stsb_multi_mt has NO Arabic split (checked on the Hub),
so component 6 uses the Arabic STS-B VALIDATION split of Omartificial-Intelligence-Space/Arabic-stsb (apache-2.0; machine-translated STS-B, 1,500 pairs; validation.csv, "score" column).
Everything else is unchanged. Data fetched to runs/retrieval/v2/data/suite/ (xnli_ar_train.parquet, ar_sarcasm_train.parquet, stsb_ar_validation.csv).

Amendment 2: alpha extension and LoRA rerun (2026-11-04 21:35)

2026-11-04 ~21:35 - AMENDMENT 2 to the pre-registered suite selection (coordinator follow-up after DRIVE14_DONE); written BEFORE any of the runs below
What happened in drive14 (facts, from the logs; the "rc=0" lines of the driver are NOT reliable because run_docker.sh masks the exit code - from now on every step is verified by the existence of its output file):
* LoRA: m3_lo_s13 trained all 615/615 steps (56 min) and then CRASHED in train_v2.py at merge_and_unload (AttributeError: 'XLMRobertaModel' has no attribute 'merge_and_unload'): my code kept the in-place-injected XLM-R module instead of the PeftModel handle. Nothing was saved (the adapters died with the process), so the LoRA dev / merges / blends / proxy2 / suite2 all failed on the missing path
within seconds. This is a bug in MY code, not a result about LoRA. proxy2 and suite2 aborted at their FIRST model (LoRA), so the freeze run and its two blends (which trained fine: dev 85.0 / 84.8 / ..., devM 78.8 / 79.7 / ...) have no proxy and suite numbers.
* select2's "PICK (interim)" was computed on the 11 complete models only.
* Tie rule: the pre-registration says "|diff| < 0.02" (strict). select code used >= best - 0.02 (inclusive). Corrected to strict; exact differences are printed to 4 decimals; the pick report states which clause decided.
DECISIONS (no re-litigation after this entry):
A. LoRA is RERUN with the identical recipe / seed (m3_lo_s13, r=32, lr 1e-4, attention+MLP, 1 epoch, seed 13) after a gated 3-minute merge self-test of the fixed code (adapter saved to disk before the merge). ~57 min. Reason: it is the only
pre-registered candidate that constrains the update DURING training (freeze did not help on devM); the GPU would otherwise idle until the pick. It enters the pool exactly as registered (raw + WiSE .75 / .5).
B. Order: (1) freeze proxy + suite → (2) select on the pool WITHOUT LoRA → winning family F by the rule as written (strict tie) → (3) ONE extra pair of alphas for F ONLY: alpha 0.30; alpha 0.35 is added ONLY if the
0.30 model fails eligibility (dev ar→en < 78.5 = bge-m3 dev ar→en 73.5 + 5.0); if 0.30 is eligible, 0.35 is NOT run; nothing below 0.30 is tried → (4) LoRA rerun + blends → (5) FINAL select on the FULL pool
(all earlier candidates + the 0.30/0.35 point(s) + LoRA models), same normalisation (pool-wide min-max, 1-point floor), same eligibility, same strict tie rule → (6) ONE MTEB(ara) scoring of that pick (= final_m3), reported even if lower.
If LoRA wins the final select there is no further alpha extension. The alphas 0.30 / 0.35 are selectable. Family names for the extension: wise → w_m3b_a30 / a35; soup → soup_a30 / a35; freeze → w_m3_fr_s13_a30 / a35.
C. Everything else in the first pre-registration (components, normalisation, eligibility, tie-break by ingredients, 3-seed reporting) is unchanged. No MTEB(ara) and no TEST number is read by any of the selection steps.

Pre-registration 3: confirmatory seeds, student rule, post-hoc curve (2026-11-04 22:10)

2026-11-04 ~22:10 - PRE-REGISTRATION 3 (coordinator additions A-D, "the user approved them all"); written BEFORE any of these runs. Order: final select → MTEB(ara) of the pick → A → D → B → C → RESULTS_v2.md → local commits (no attribution).
A. CONFIRMATORY REPLICATION (the confirmatory numbers of the report; reported whatever they show).
* Recipe FROZEN = the (family, alpha) of results/suite_pick.json from the final select. Nothing about it is changed after this line.
* Two NEW seeds, 16 and 17, never used in any selection step or training run so far (used: 13, 14, 15 for the m3_b recipe). Identical training arguments to the pick's family (wise/soup: the m3_b recipe = bge-m3, v2 data, same:1 negatives, 1 epoch, lr 1e-5, scale 50, CachedMNRL mini-batch 16, exclude file; freeze / lora: the registered freeze / LoRA arguments),
identical WiSE alpha towards bge-m3 (no soup unless the pick is a soup: then seeds 16, 17 AND 18 are trained and averaged into ONE confirmatory soup, a stated deviation from "2 seeds").

* Report per seed and for the pair: TEST (4 directions + macro, paired bootstrap CIs vs bge-m3 and vs v1), mixed-index diagnostic, and MTEB(ara) Mean(Task) (all 19 tasks) for each. No dev or suite number is used to choose between them.
D. STUDENTS from the final teacher (teacher embeddings tag m3f re-computed from final_m3): KD-only, 2 epochs, seeds 13/14/15, e5-base and e5-small. FINAL STUDENT RULE (registered now):
final_base / final_small = the uniform 3-seed soup of the students, unless the soup's dev macro is more than 0.5 below the MEAN dev macro of its three seeds (failed averaging), in which case seed 13.
Then MTEB(ara) once for each (final_base, final_small) against e5-base 67.888 / e5-small's own cached score. Reported even if lower.
B. BLEND CURVE (POST-HOC FIGURE, NOT USED FOR SELECTION): m3_b s13 family at alpha in {0, 0.3, 0.4, 0.5, 0.6, 0.75, 1.0}: TEST ar→en, TEST macro, devM, mMARCO proxy, guards MIRACL-HN.v2 (ar) and NamaaMrTydiReranking.
Existing results are reused (bge-m3 = alpha 0, m3_b_s13 = alpha 1, wm_a75 / final_m3_wa75_s13, the pick); only missing cheap evals are run (TEST of the blends; MTEB guards only, no other MTEB tasks per alpha). → results/alpha_curve.json.
C. MTEB(ara) UNCERTAINTY (CPU only, from the result caches): final_m3 vs bge-m3, the students vs e5-base / e5-small, the confirmatory seeds vs bge-m3: per-task win / loss / tie counts,
bootstrap CI of the Mean(Task) difference by resampling the 19 tasks (10,000 resamples, seed 13), exact two-sided paired sign test. → results/mteb_paired.json.
Infrastructure: run_docker.sh is NOT touched while this queue runs (coordinator fixes its exit-code masking afterwards); my queue reports completion by DRIVE16_DONE in logs/queues.done.

Recipe frozen, suite pick, start of the confirmatory seeds (2026-11-04 22:46) - 22:46

RECIPE FROZEN at 2026-11-04T22:46:54+03:00 (before any confirmatory seed is trained): family wise, alpha 0.4, pick w_m3b_a40, per-seed file recipe name w_m3b_a40. Seeds 16/17 are trained only after this line.
- 22:46 SUITE PICK (pre-registered rule, full pool): w_m3b_a40 (family wise, alpha 0.4). top score soup_a40 0.7612; runner-up w_m3b_a40 0.7486; difference 0.0127 → within the tie band (< 0.02) among ['soup_a40', 'w_m3b_a40'] → decided by the INGREDIENTS clause. Saved as runs/retrieval/v2/models/final_m3 (clean sentence-transformers folder). ONE MTEB(ara) scoring now; reported whatever it is.
- 23:13 CONFIRMATORY REPLICATION starts (recipe frozen at 2026-11-04T22:46:54+03:00): frozen recipe = family wise, alpha 0.4 (pick w_m3b_a40); raw run name prefix m3_b; NEW seeds: 16 17 (never used in any selection or training so far). Training arguments: –base bge-m3 –data v2 –negs same:1 –epochs 1 –lr 1e-5 –scale 50 –loss cached –mini-batch 16 –eval-bs 32 –exclude /lab/runs/retrieval/v2/data/exclude_qids.json –test –keep-model

Correction of the contamination labels (exact copies) (2026-11-05 07:2x) - 2026-11-05 07:2x

CORRECTION (coordinator's read-only checks exact_overlap.py / exact_leak.py): (1) guard_split labelled queries with the Jaccard detector only; its inverted index skips words with more than 300 postings, so short common-word questions got no candidates. MIRACL-HN.v2 ar dev: 893/1000 queries are EXACT normalised copies of TyDi-ar train questions, the detector flagged 814 (816 with MIRACL train). The labels now also flag exact normalised copies; the report is recomputed on CPU from the cached per-query scores (no GPU). (2) The "42% verbatim" statement in the 05:40 entry was the 408/1000 NEAR-DUPLICATES against v1's training pool, not verbatim copies of the full TyDi train set; the docstrings of contam_v2.py and guard_split.py are corrected. (3) After exclude_qids, 9 kept training rows (8 distinct questions, all slice train_tydi_ar) equal MIRACL-HN.v2 ar dev queries exactly (0.8% of 998 normalised eval queries); NamaaMrTydi: 0. Not retrained; disclosed in RESULTS_v2.md section 2 / 8 / 11 and in the three model cards. New read-only script lab/retrieval/v2/leak_check.py → results/leak_exact.json (also counts the translated copies via src_qid and the effect on MIRACL-HN without the leaked queries).